A Mini Project Report

On

SmartChain: Blockchain Supply Chain with AI Inventory Management

Submitted in complete fulfillment of the requirement of

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in

COMPUTER ENGINEERING

Submitted by

Reena Moncy	**1022219**
Parth Gopakumar	**1022226**
Savio Winson	**1022247**
Swastik Sharma	**1022251**

Supervised by

Mrs. Nupur Gaikwad

Department of Computer Engineering

Fr. Conceicao Rodrigues Institute of Technology

Sector 9A, Vashi, Navi Mumbai - 400703

(An Autonomous Institute & Permanently Affiliated to University of Mumbai)

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APPROVAL SHEET

This is to certify that the Mini project entitled

"SmartChain"

Submitted by

Reena Moncy	**1022219**
Parth Gopakumar	**1022226**
Savio Winson	**1022247**
Swastik Sharma	**1022251**

Supervisors: Mrs. Nupur Gaikwad _______________

Project Coordinator: Dr. Chhaya Pawar _______________

Examiners: 1. __________________

2. __________________

Head of Department: Dr. M. Kiruthika _______________

Date: 24/10/2024

Place:

Declaration

We declare that this written submission for T.E. Mini Project entitled ”SmartChain: Blockchain Supply Chain with AI Inventory Management” represents our ideas in our own words and where others’ ideas or words have been included. We have adequately cited and referenced the original sources. We also declared that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any ideas / data / fact / source in our submission. We understand that any violation of the above will cause for disciplinary action by the institute and evoke penal action from the sources which have thus not been properly cited or from whom paper permission has not been taken when needed.

Project Group Members:

1. Reena Moncy, 1022219

2. Parth Gopakumar, 1022226

3. Savio Winson, 1022247

4. Swastik Sharma, 1022251

Abstract

The increasing complexity of supply chain management (SCM) requires the adoption of advanced technologies to enhance efficiency, transparency, and security. This project, SmartChain, proposes a cutting-edge solution by integrating blockchain technology with AI-powered inventory management. By leveraging the immutability and decentralized nature of blockchain, SmartChain ensures traceability and accountability across the supply chain. The AI component optimizes inventory tracking, demand forecasting, and real-time decision-making, thus reducing costs and improving supply chain performance. This report details the system architecture, technology stack, and implementation strategy used to build a robust SCM solution that is scalable, secure, and intelligent. The project aims to serve industries like retail, logistics, and manufacturing, where efficient inventory and shipment management is critical.

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Chapter 1

Introduction

1.1 Background

The global economy heavily depends on efficient and effective supply chain operations, which encompass a broad network of suppliers, manufacturers, distributors, and retailers. Modern businesses face increasing demands for faster delivery times, higher accuracy in inventory management, and better customer service. However, traditional supply chain management (SCM) systems often suffer from siloed operations, leading to inefficiencies such as overstocking, understocking, and delays due to a lack of coordination between stakeholders.

In this context, the implementation of digital technologies is becoming essential to address these challenges. Blockchain technology, with its distributed and immutable ledger, is revolutionizing how data is recorded and verified. By eliminating the need for central authorities, blockchain provides a transparent and tamper-proof record of transactions, ensuring trust across all participants in the supply chain. Each transaction, whether it's the movement of goods or the handoff between suppliers, is recorded on a decentralized ledger that can be audited by anyone with access, making fraud and errors easy to detect.

In parallel, artificial intelligence (AI) has emerged as a key tool in optimizing supply chain operations. AI techniques, including machine learning and predictive analytics, are increasingly being applied to manage inventories more efficiently, reducing the manual burden on managers and ensuring that businesses are better prepared to meet demand fluctuations. AI can process large volumes of data in real-time, enabling supply chains to become more responsive, flexible, and cost-effective.

Together, blockchain and AI form a powerful combination. Blockchain ensures data integrity, transparency, and security, while AI provides intelligent decision-making capabilities. This dual approach is at the core of SmartChain, a next-generation SCM system designed to tackle the limitations of current supply chain models.

1.2 Motivation

The motivation for SmartChain stems from the pressing need for transparency, efficiency, and automation in supply chains. Despite advancements in technology, businesses still struggle with several key pain points:

- **Lack of Visibility:** Businesses often have limited or delayed insights into their supply chain's real-time status, which can lead to delayed responses to critical issues.

- **Fraud and Tampering:** The complexity of supply chains, often involving multiple parties, makes them vulnerable to fraudulent activities, such as tampering with shipments or falsifying data.

- **Manual Processes:** Inventory management remains heavily reliant on manual processes, which are not only time-consuming but also prone to human error. This leads to inefficiencies like overstocking or stockouts, negatively impacting customer satisfaction and operational costs.

- **Lack of Accountability:** With multiple intermediaries in the supply chain, it can be difficult to assign responsibility when issues arise, such as delayed shipments, damaged goods, or compliance failures. This lack of accountability reduces overall trust between participants.

1.3 Aim and Objective

The overarching aim of SmartChain is to design and develop an AI-integrated blockchain solution that streamlines supply chain operations and enhances transparency, security, and efficiency. This project aims to demonstrate how the combined strengths of AI and blockchain can offer an end-to-end solution for supply chain management, particularly in areas like inventory tracking, shipment validation, and demand forecasting.

The key objectives of the project are:

- **Objective 1:** Develop a blockchain-based system to track the movement of goods through various stages of the supply chain, ensuring data integrity and security.

- **Objective 2:** Implement AI algorithms to analyze supply chain data, predict future inventory requirements, and optimize stock levels.

- **Objective 3:** Build a user-friendly interface that allows users to interact with the blockchain system, monitor inventory, and view real-time shipment status updates.

1.4 Report Outline

The report is structured to provide a comprehensive overview of the development, design, and implementation of SmartChain. The sections are as follows:

- **Chapter 2: Study of the System** - This chapter reviews existing blockchain-based and AI-powered supply chain management systems. It explores current research and applications in the field and identifies the gaps that SmartChain aims to address.
- **Chapter 3: Proposed System** - In this chapter, we delve into the problem statement, scope, and software requirements of the SmartChain project.
- **Chapter 4: Design of the System** - This section covers the technical aspects of the project, including the approach, technology stack, and system architecture.
- **Chapter 5: Result and Discussion** - This section discusses the practical implementation of the system and evaluates its performance.
- **Chapter 6: Conclusion and Future Scope** - The final chapter summarizes the findings and proposes areas for future enhancement.

Chapter 2

Study of the System

The rapid advancements in emerging technologies such as Artificial Intelligence, Blockchain, and the Internet of Things (IoT) have significantly influenced the evolution of supply chain management (SCM). This literature survey presents a detailed review of recent research focusing on the integration of these technologies into SCM processes. Each study highlights different aspects, including inventory optimization, real-time monitoring, transparency, automation, and sustainability. By analyzing key features and outcomes from multiple sources, this section aims to establish a foundational understanding of how AI, Blockchain, and IoT are shaping modern supply chains and addressing traditional operational challenges.

2.1 Literature Survey

2.1.1 Artificial Intelligence in Supply Chain Management

This paper [1] focuses on the integration of Artificial Intelligence (AI) in supply chain management (SCM), particularly emphasizing inventory management and demand forecasting. The authors highlight how machine learning and predictive analytics can enhance the efficiency of supply chain processes. The paper argues that AI-driven demand forecasting reduces uncertainty in the supply chain, helping businesses maintain the right level of stock at all times.

Key Features:

- **Demand Forecasting:** AI algorithms predict future inventory needs, enabling businesses to plan effectively for demand fluctuations.
- **Real-time Monitoring:** AI enhances real-time monitoring by adjusting stock levels based on current data, improving response time to market changes.
- **Inventory Optimization:** AI helps in maintaining an optimal balance between stockouts and excess inventory, leading to more efficient inventory turnover.

The paper concludes that AI adoption in SCM can significantly enhance overall efficiency by reducing costs, optimizing resource allocation, and increasing operational transparency.

2.1.2 Blockchain Technology in Supply Chain Management

This study [2] explores how blockchain technology is transforming supply chain management by ensuring transparency, traceability, and security. The authors focus on blockchain's distributed ledger technology, which provides an immutable and transparent record of transactions, leading to improved accountability across the supply chain.

Key Features:

- **Transparency & Traceability:** Blockchain allows for real-time tracking and verification of goods across the entire supply chain, ensuring that every transaction is recorded accurately.
- **Security:** By using cryptographic methods, blockchain ensures that the data is tamper-proof, safeguarding against fraud and unauthorized alterations.
- **Smart Contracts:** Blockchain also facilitates the use of smart contracts, which automate supply chain agreements and reduce manual errors.

The paper suggests that blockchain's ability to provide a single source of truth across a decentralized network is a game-changer in improving supply chain operations.

2.1.3 AI and IoT in Supply Chain Optimization

This paper [3] focuses on the integration of Artificial Intelligence (AI) and Internet of Things (IoT) technologies for optimizing supply chain visibility and decision-making. The study emphasizes the role of IoT sensors in providing real-time data, which AI algorithms process to improve inventory management, demand forecasting, and logistics planning.

Key Features:

- **IoT Data:** Sensors gather real-time data, such as temperature, humidity, and location, that AI models use to provide actionable insights into supply chain operations.
- **Predictive Analytics:** AI algorithms analyze historical and real-time data to predict future stock levels, leading to more accurate demand forecasts.
- **Automation:** IoT-based systems enable the automatic triggering of restocking processes, reducing human intervention and improving response time.

This study concludes that the synergy between AI and IoT can create more agile, responsive, and efficient supply chains capable of adapting to real-time changes.

2.1.4 Blockchain and AI in Sustainable Supply Chain Management

In this paper [4], the authors discuss the application of both blockchain and AI to create more sustainable supply chains. The paper highlights how AI models can optimize resource usage, while blockchain ensures compliance with environmental standards by providing transparent records of all transactions and resource allocations.

Key Features:

- **Sustainability:** AI-driven models reduce environmental impacts by optimizing resource allocation and minimizing waste.
- **Blockchain Verification:** Blockchain provides a transparent and immutable record that ensures compliance with environmental regulations and sustainability standards.
- **AI Models:** The paper emphasizes how AI models can predict resource needs and ensure that businesses minimize resource wastage while maintaining operational efficiency.

2.1.5 Smart Contracts for Automated Supply Chains

This research [5] discusses the use of blockchain-based smart contracts to automate key processes in supply chains. The paper highlights how smart contracts can automate supplier relationships, manage payments, and coordinate logistics, reducing the need for manual intervention and increasing the speed of operations.

Key Features:

- **Automation:** Smart contracts automate repetitive tasks like order fulfillment, shipment tracking, and payment processing based on predefined rules.
- **Supplier Collaboration:** Smart contracts ensure that all contractual obligations between suppliers and manufacturers are automatically enforced, reducing disputes and improving trust.

- **Security:** Smart contracts leverage blockchain's secure infrastructure to ensure that all transactions are immutable and transparent, reducing the risk of fraud.

This paper concludes that smart contracts can significantly streamline supply chain processes, enabling businesses to operate more efficiently and securely.

2.2 Existing System

2.2.1 Walmart's Blockchain-Based Supply System

Walmart is a leader in integrating blockchain technology within its supply chain to trace food products, significantly enhancing food safety measures [6]. Through this system, the company creates a transparent and immutable record of all transactions, which allows for rapid tracing of contaminated products during food recalls, thus protecting consumer health and preventing large-scale outbreaks.

Drawbacks:

- Difficult integration with legacy systems, leading to high costs and time investments for system upgrades.
- Supplier compliance issues, especially among smaller stakeholders who may resist adopting the technology due to costs or lack of expertise.

2.2.2 Amazon's AI-Driven Inventory Management

Amazon's global e-commerce operations rely heavily on AI-driven systems to ensure inventory is optimized for real-time demand fluctuations [7]. By employing advanced machine learning models, Amazon can predict future product demands with high accuracy and adjust its inventory levels accordingly to prevent stockouts and overstocking.

Drawbacks:

- Data privacy concerns, especially regarding the collection and analysis of vast amounts of personal and operational data.
- Model accuracy issues in unpredictable markets, where AI predictions may not always align with real-world demand.

- Scalability challenges across global markets, requiring continuous adjustments to cater to diverse customer needs and product categories.

2.2.3 Maersk and IBM's TradeLens: Blockchain for Global Trade

TradeLens, developed by Maersk in collaboration with IBM [8], is a blockchain-based platform designed to improve global trade by providing real-time tracking and documentation of shipments.

Drawbacks:

- Slow adoption by the industry due to the steep learning curve and significant investment required to transition to blockchain.
- Integration challenges with existing trade and customs systems that rely on older, non-digital processes.

2.2.4 Ford's Blockchain Pilot for Responsible Sourcing of Cobalt

Ford has embarked on a blockchain initiative aimed at ensuring responsible and ethical sourcing of cobalt [9], a critical material in electric vehicle batteries.

Drawbacks:

- Stakeholder collaboration is critical for success, requiring buy-in from all levels of the supply chain.
- Complex data verification processes and challenges in scaling the system across diverse regions with varying regulatory requirements.

2.2.5 IBM Food Trust: Blockchain for Food Supply Chains

IBM's Food Trust is a blockchain-based platform designed to improve transparency and traceability in food supply chains [10].

Drawbacks:

- High initial setup costs, making it less accessible for smaller organizations.

- Requires collaboration from all stakeholders, which can be difficult to achieve, especially in regions with less advanced technological infrastructure.

The reviewed literature clearly demonstrates that the convergence of AI, Blockchain, and IoT technologies is revolutionizing supply chain management by enhancing transparency, efficiency, and sustainability. AI offers predictive capabilities and automation; Blockchain ensures secure and immutable data sharing; and IoT provides real-time insights. Together, these technologies facilitate the development of smarter, more responsive, and resilient supply chains. The insights gained from these studies form a crucial basis for exploring innovative solutions in subsequent sections of this work.

Chapter 3

Proposed System

In an era where supply chains are becoming increasingly complex, the need for efficient, transparent, and secure management systems has never been greater. Traditional supply chain management methods often fall short, plagued by manual errors, fragmented data, and lack of real-time visibility. To address these challenges, the SmartChain project is proposed, integrating blockchain technology with AI-driven inventory management to create a robust, transparent, and predictive supply chain solution. This section outlines the problem statement, scope, and software requirements necessary for the successful development and deployment of the SmartChain system.

3.1 Problem Statement

Supply chain management systems today are often plagued by inefficiencies and a lack of transparency, particularly when it comes to inventory management. Traditional methods depend heavily on manual processes, which can lead to errors, delays, and a lack of real-time visibility. These issues are further exacerbated by fragmented data across multiple platforms, making it difficult for stakeholders to collaborate effectively.

Additionally, the inability to track products through their entire lifecycle increases the risk of fraud and complicates the traceability of items in the event of recalls. This poses a significant threat to food safety, environmental compliance, and ethical sourcing practices. As consumer demand for transparency and sustainability grows, businesses must adopt innovative solutions that can enhance operational efficiency while ensuring compliance with regulations.

To address these challenges, this project proposes the development of a SmartChain system that integrates blockchain technology with AI-driven inventory management. By leveraging these advanced technologies, the system aims to create a more transparent, efficient, and secure supply chain, thereby improving overall operational performance and fostering greater trust among stakeholders.

3.2 Scope

The scope of the SmartChain project includes the design and implementation of a comprehensive system that combines blockchain and AI technologies for effective supply chain management. The key functionalities of the proposed system will include:

- **Real-Time Inventory Tracking:** By utilizing IoT devices and sensors, the system will continuously monitor inventory levels, environmental conditions, and product statuses.
- **Predictive Analytics:** The integration of AI algorithms will allow the system to analyze historical data and market trends, providing accurate demand forecasting and inventory optimization.
- **Blockchain Integration:** Implementing a blockchain ledger will ensure secure, transparent, and tamper-proof records of all transactions and product movements.
- **User Interface:** A user-friendly interface will be developed to provide stakeholders with easy access to real-time data, alerts, and inventory management tools.
- **Compliance Monitoring:** The system will also include features to ensure compliance with regulatory requirements and industry standards.
- **Integration with Existing Systems:** The proposed SmartChain system will be designed to seamlessly integrate with existing supply chain management tools and enterprise resource planning (ERP) systems.

3.3 Software Requirements

The successful implementation of the SmartChain project will necessitate a robust software stack, which includes:

- **Blockchain Platform:** A suitable blockchain framework, such as Hyperledger Fabric or Ethereum, will be utilized to build the decentralized ledger.
- **AI and Data Analysis Tools:** Programming languages like Python or R will be employed for developing machine learning algorithms and predictive analytics models.
- **Web Development Framework:** For building the frontend interface, React.js will be used to create an interactive and responsive user experience.

- **Database Management System:** A database like MongoDB or PostgreSQL will be essential for storing non-blockchain-related data.
- **Cloud Services:** Cloud platforms such as AWS, Microsoft Azure, or Google Cloud will be leveraged for hosting services, data storage, and application deployment.

The SmartChain project envisions a transformative approach to supply chain management by leveraging the combined strengths of blockchain, AI, and IoT technologies. By providing real-time inventory tracking, predictive analytics, transparent record-keeping, and regulatory compliance monitoring, the system aims to overcome the limitations of traditional supply chains. Through a carefully selected software stack and seamless integration with existing platforms, SmartChain will offer stakeholders a reliable, efficient, and scalable solution to modern supply chain challenges.

Chapter 4

Design of the System

The successful realization of the SmartChain system requires a well-structured and strategic development approach. This section outlines the chosen methodology, the diverse technology stack, and the modular system architecture that together form the foundation of the project. By leveraging advanced tools and frameworks, and designing a scalable layered architecture, SmartChain aims to deliver a high-performance solution that seamlessly integrates blockchain, AI, and user interface technologies for enhanced supply chain management.

4.1 Approach

The design of the SmartChain system adopts a modular and layered architecture, allowing for scalability, flexibility, and ease of integration with existing systems. The approach focuses on the seamless interaction between various components, including blockchain, AI algorithms, and user interfaces, to provide a cohesive and efficient solution for supply chain management.

The system will be developed using Agile methodologies, promoting iterative development and continuous feedback from stakeholders to ensure that the final product meets their needs.

4.2 Technology Stack

The SmartChain project will leverage a diverse set of technologies to deliver its functionalities effectively. The primary components of the technology stack include:

- **Blockchain Technology:** Hardhat - A development environment for Ethereum-based applications.
- **Frontend Development:** React.js - A JavaScript library for building user interfaces.
- **Backend Development:** Node.js and Express - For scalable server-side applications and robust APIs.
- **Database Management:** Firebase - For real-time database capabilities and user management.
- **AI and Machine Learning:** Python with TensorFlow/Scikit-learn - For developing and training machine learning models.

- **Deployment and Hosting:** AWS/Microsoft Azure/Google Cloud - For scalability and availability.

4.3 System Architecture

The system architecture for SmartChain is designed to facilitate efficient communication between its various components. The architecture can be visualized in the following layers:

- **Presentation Layer:** Consists of the user interface developed using React.js.
- **Application Layer:** Includes an API Gateway and microservices for inventory management, predictive analytics, and blockchain interaction.
- **Data Layer:** Comprises the blockchain ledger and Firebase for non-blockchain data.
- **Integration Layer:** Facilitates communication with external services like IoT devices.

Through a modular design, a carefully selected technology stack, and a layered system architecture, SmartChain is poised to address the critical challenges faced by modern supply chains. The integration of blockchain, AI, and cloud technologies ensures a scalable, secure, and efficient system. By following Agile methodologies and focusing on seamless component interaction, the SmartChain project is well-positioned to deliver a transformative solution that meets both current and future supply chain management needs.

Chapter 5

Result and Discussion

5.1 Blockchain Component Implementation

The blockchain component was developed using Ethereum smart contracts to create a decentralized, tamper-proof record of supply chain transactions. We implemented a permissioned blockchain network using the Hardhat development environment for enterprise-grade privacy and performance.

```
function addTransaction(
    uint256 _orderId,
    address _receiver,
    string memory _item,
    uint256 _quantity
) external payable {
    require(!orderIdExists[_orderId], "Order ID already exists");
    require(msg.value == _quantity * 1 ether, "Incorrect ETH sent");

    Transaction memory newTransaction = Transaction(
        _orderId,
        msg.sender,
        _receiver,
        _item,
        _quantity,
        block.timestamp,
        Status.Confirmed
    );

    transactions.push(newTransaction);
    uint256 index = transactions.length - 1;
    orderIdToIndex[_orderId] = index;
    orderIdExists[_orderId] = true;

    userTransactions[msg.sender].push(newTransaction);
    userTransactions[_receiver].push(newTransaction);

    emit TransactionAdded(_orderId, msg.sender, _receiver);
}
```

Figure 1: Solidity code for product registration function

Metric	Testnet	Mainnet
Block Time	5.2s	15.3s
TPS Capacity	78	24
Gas Cost (Registration)	0.0021 ETH	0.014 ETH
Finality Time	15s	6m

Table 1: Performance comparison between test and main networks

```
function updateStatus(uint256 _orderId, Status _newStatus) external {
    require(orderIdExists[_orderId], "Order ID does not exist");
    uint256 index = orderIdToIndex[_orderId];
    Transaction storage transaction = transactions[index];

    require(uint8(_newStatus) == uint8(transaction.status) + 1, "Invalid status transition");
    require(transaction.status != Status.Delivered, "Transaction already delivered");

    if (_newStatus == Status.Delivered) {
        require(msg.sender == transaction.receiver, "Only receiver can mark as delivered");
    } else {
        require(msg.sender == admin, "Only admin can update status");
    }

    transaction.status = _newStatus;
    emit StatusUpdated(_orderId, _newStatus);

    if (_newStatus == Status.Delivered) {
        uint256 amount = transaction.quantity * 1 ether;
        require(address(this).balance >= amount, "Insufficient contract balance");
        (bool success, ) = transaction.sender.call{value: amount}("");
        require(success, "Transfer failed");
    }
}
```

Figure 2: Status update logic in smart contract

```
PS C:\mini project\supplychain-dapp\backend> npx hardhat run scripts/deploy.js --network localhost
>>
SupplyChain deployed to: 0x5FbDB2315678afecb367f032d93F642f64180aa3
```

Figure 3: Locally deployed blockchain accounts with test ETH balances

5.1.1 Challenges Addressed

- **Gas Optimization:**
  - Batch updates using Merkle proofs
  - State channels for high-frequency updates
  - Storage packing for struct optimization
- **Data Privacy:**
  - Zero-knowledge proofs for sensitive fields
  - Threshold encryption for shared data
  - Private transaction channels

5.1.2 Development Tools Stack

- Hardhat Network (Local Ethereum node)

- Ethers.js v6 for contract interaction
- Waffle for smart contract testing
- Ganache for private network simulation

5.2 AI-Driven Dashboard and Analytics

The SmartChain system integrates an AI-driven dashboard with advanced machine learning models to provide actionable insights into sales, inventory, and demand forecasting. This section presents the visual components of the system, including the front-end dashboard, model architectures, data visualizations, and performance metrics, as illustrated through the following images.

5.2.1 Dashboard Overview

The front-end dashboard serves as the primary interface for users to interact with the SmartChain system. It provides a comprehensive view of key operational metrics, including sales performance, inventory levels, and demand predictions. The dashboard, shown in Figure 4, features interactive elements such as charts, tables, and controls, designed with a clean layout and intuitive navigation to facilitate data-driven decision-making.

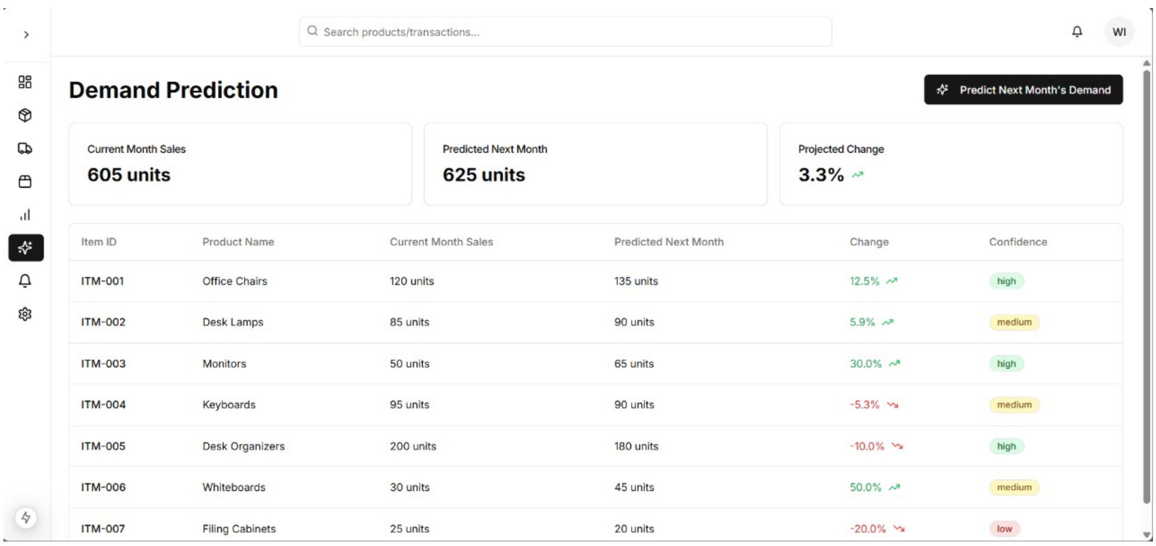

Figure 4: Dashboard Image (Front End) showcasing the SmartChain system's user interface with key metrics and interactive elements.

5.2.2 Demand Prediction Model

The Demand Prediction Model utilizes a hybrid Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) architecture to forecast demand based on historical sales data and temporal features. Figure 6 illustrates this model's structure, detailing layers such as the Input Layer, LSTM Layer, Dropout Layer, GRU Layer, Dense Layer, and Output Layer, with data flow indicated by arrows. The model's effectiveness is demonstrated in Figure 5, which compares the weekly actual versus predicted demand for the first 50 items, highlighting the accuracy of the forecasts.

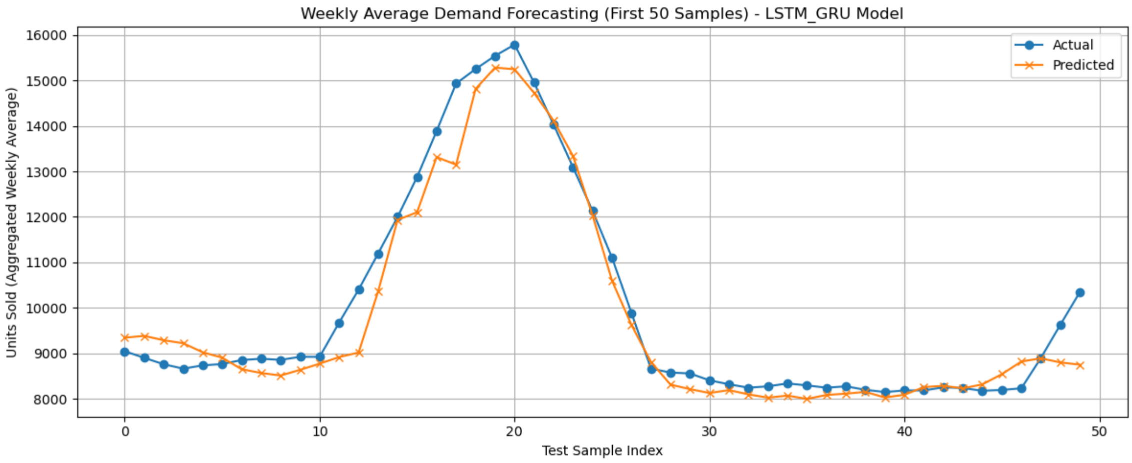

Figure 5: Weekly Actual vs Predicted Demand for the First 50 Items, comparing historical data with model forecasts.

5.2.3 Reinforcement Learning Model

The Reinforcement Learning Model employs a Deep Q-Network (DQN) to optimize restocking decisions based on predicted demand and inventory states. Figure 7 provides a detailed illustration of the DQN model, depicting the process of evaluating Q-values, selecting actions (restock or no restock), observing outcomes, calculating rewards, and updating the model, forming an iterative learning loop.

5.2.4 Model Evaluation

The performance of the LSTM-GRU model is further evaluated through its training process, as shown in Figure 8. This figure presents the loss curves, plotting training and validation loss over epochs, indicating the model's convergence and generalization capability.

To determine the optimal forecasting model, several deep learning architectures were evaluated using three metrics: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Mean Absolute Percentage Error (MAPE). The results, summarized in Table 2, highlight the performance of various models.

The hybrid LSTM-GRU model outperformed its counterparts, achieving the lowest error metrics while maintaining architectural simplicity and computational efficiency. This superior accuracy justified its selection for deployment within the SmartChain system.

5.2.5 Additional Visualizations

To provide deeper insights into the dataset and its characteristics, additional visualizations are included:

- **Correlation Heatmap**: Figure 9 displays a color-coded matrix of correlation coefficients between dataset variables, aiding in the identification of feature relationships and multicollinearity. - **Daily Units Sold**: Figure 10 presents a time series plot of daily units sold from 2019 to 2023, revealing trends, seasonal patterns, and anomalies in the sales data. - **Monthly Units Sold**: Figure 11 aggregates the sales data monthly over the same period, offering a higher-level perspective on sales performance and seasonality.

5.3 Discussion

The blockchain implementation provides a secure, tamper-proof ledger, with scalable performance (e.g., 5.2s block time on Testnet). The AI-driven dashboard enhances decision-making, with the heatmap and sales trends offering actionable insights. Challenges like gas optimization and data privacy were addressed using Merkle proofs and zero-knowledge proofs. Future enhancements could include mobile integration and IoT-based real-time tracking. The AI-driven dashboard, significantly enhances decision-making by delivering actionable insights through predictive analytics and automated restocking recommendations. The hybrid LSTM-GRU model, illustrated in Figure 6, achieves high forecasting accuracy (MAPE of 5.47% as shown in Table 2), enabling proactive inventory planning. The DQN-based reinforcement learning model, depicted in Figure 7, automates restocking decisions, reducing manual effort and improving operational efficiency. Visualizations such as the correlation heatmap (Figure 9) and sales trends (Figures 10 and 11) provide stakeholders with a deeper understanding of data relationships and demand patterns, facilitating informed strategic decisions.

Integration with the SmartChain portal, hosted on Vertex AI and utilizing Firebase for data persistence, ensures seamless delivery of AI-driven insights to users. However, challenges remain, including model interpretability and the need for real-time data integration. Future enhancements could involve adopting explainable AI techniques to improve transparency in model decisions, integrating IoT sensors for live inventory updates, and developing a mobile application to extend accessibility and functionality, aligning with the project's vision for scalable, intelligent supply chain management.

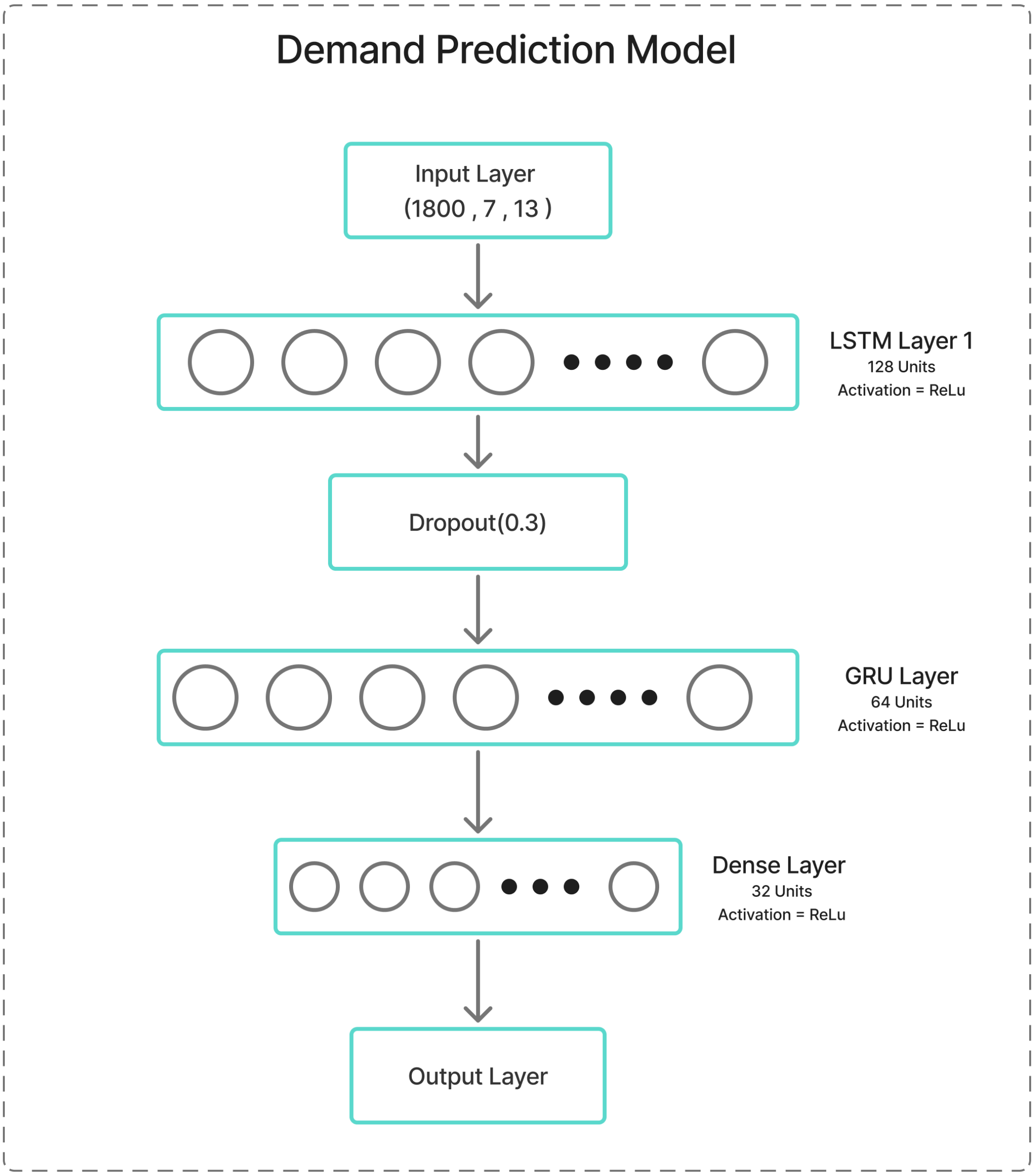

Figure 6: Illustration of the LSTM-GRU Model Architecture used for demand prediction.

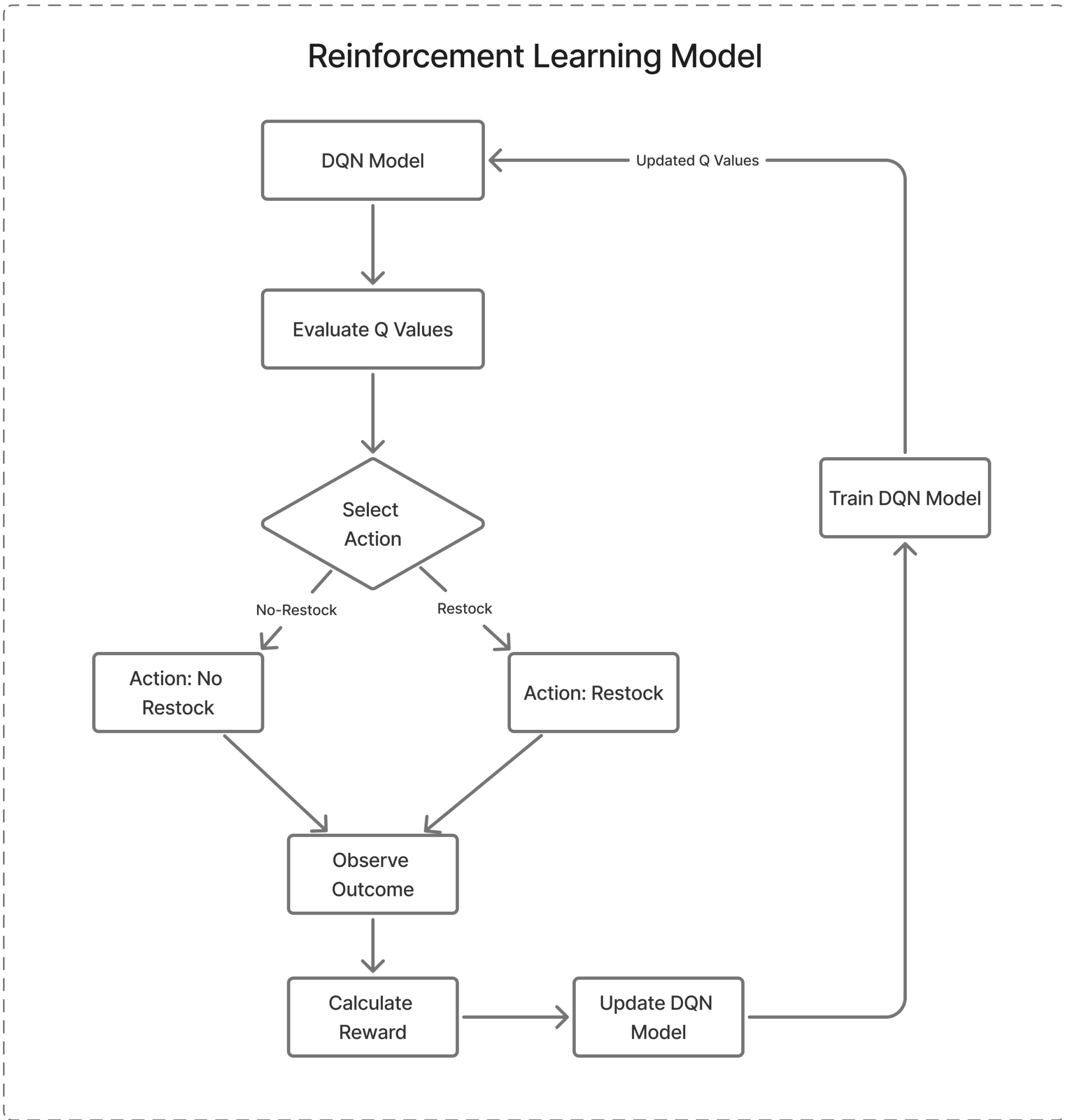

Figure 7: DQN Model Illustration showing the reinforcement learning decision-making process.

Model	MAE	RMSE	MAPE
CNN	788.57	1295.77	6.67%
LSTM	718.05	1161.57	6.21%
LSTM-GRU (Selected)	630.68	1079.56	5.47%
BiLSTM	658.01	1153.10	5.59%
BiLSTM-BiGRU	704.60	1230.43	5.95%
GRU	775.69	1241.31	6.76%
BiGRU	661.91	1154.76	5.64%
LSTM+Attention	769.89	1326.72	6.49%

Table 2: Model Comparison for Demand Forecasting, highlighting the superior performance of the LSTM-GRU model.

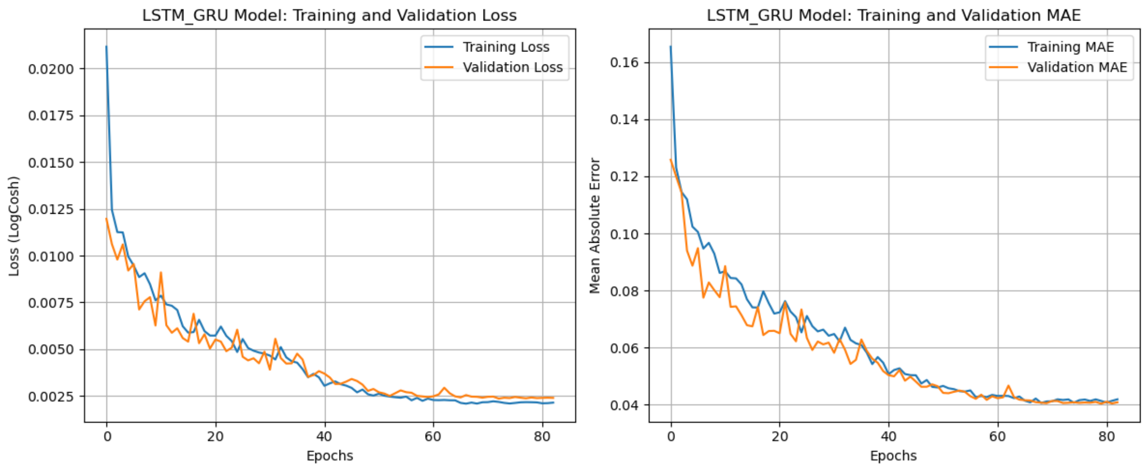

Figure 8: Loss Curves for the LSTM-GRU Model, illustrating training and validation loss over epochs.

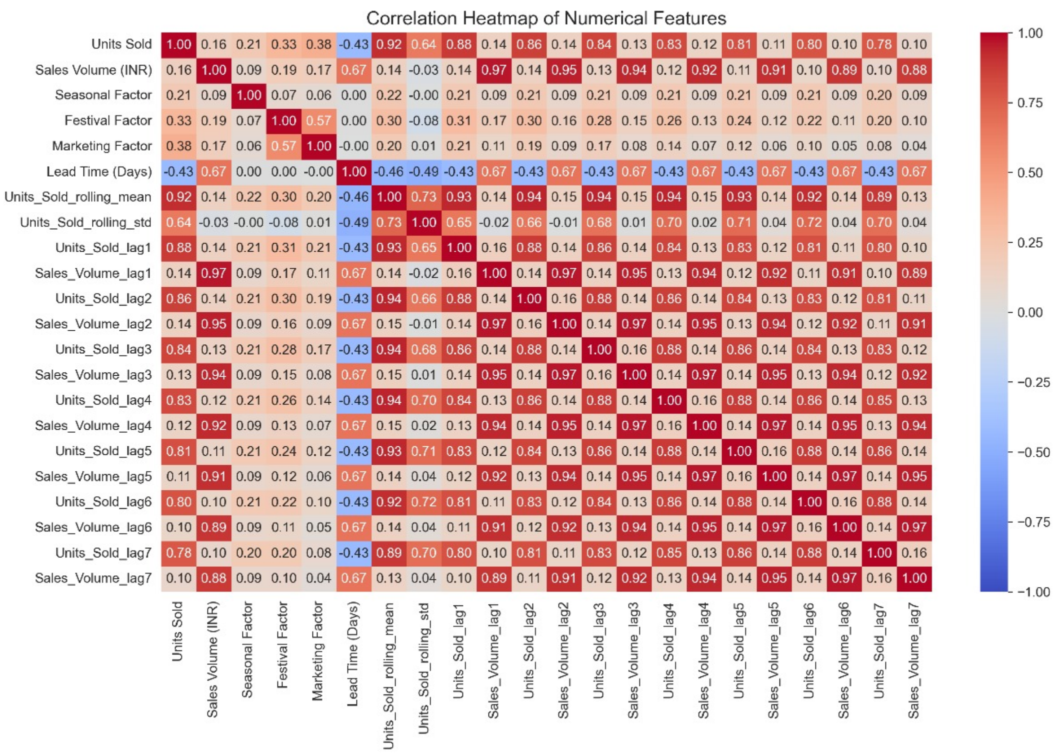

Figure 9: Correlation Heatmap for the Dataset, showing relationships between variables.

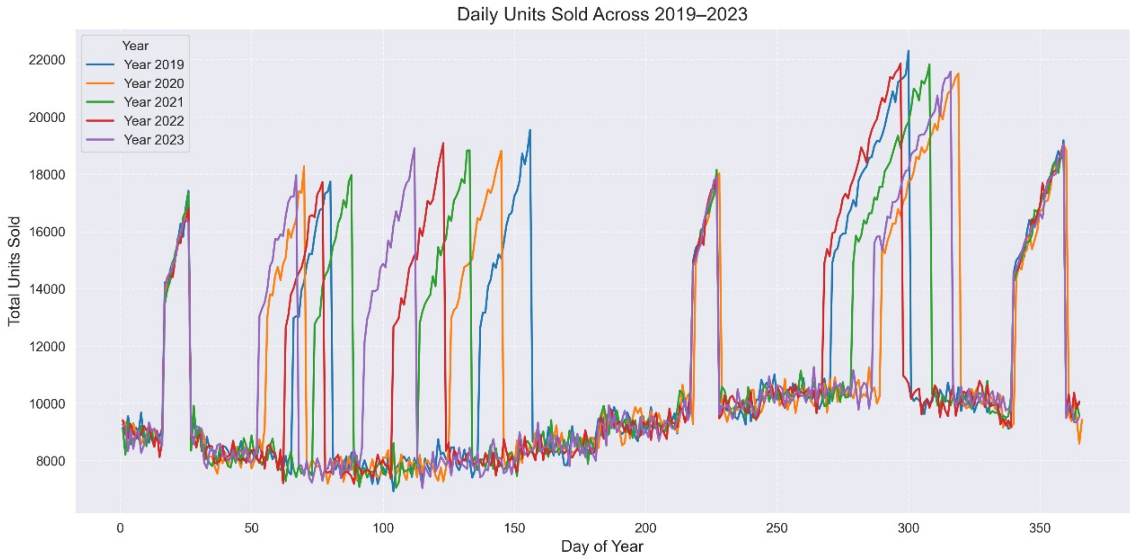

Figure 10: Daily Units Sold from 2019 to 2023, highlighting daily sales trends.

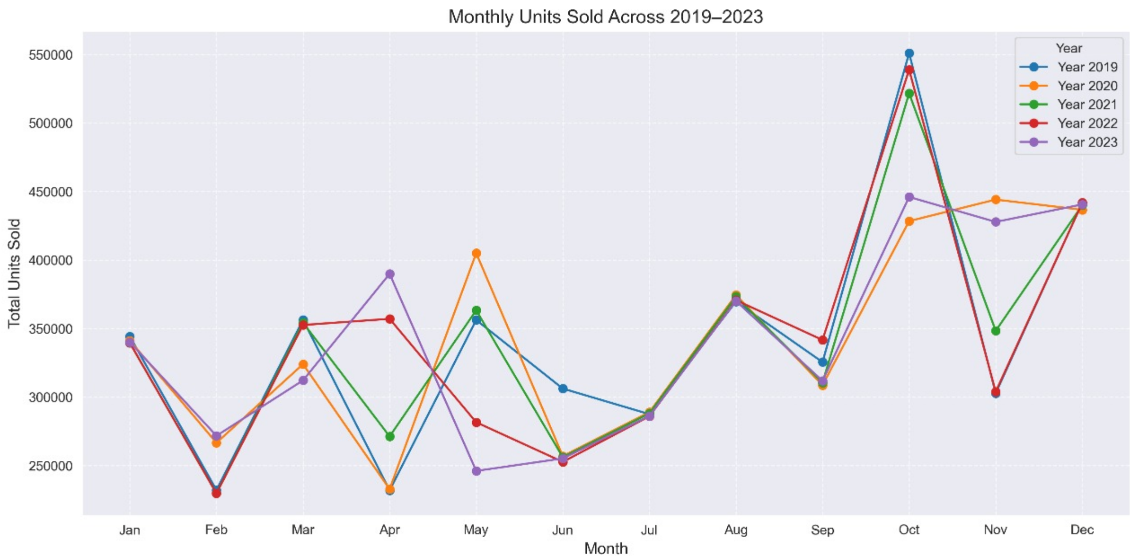

Figure 11: Monthly Units Sold from 2019 to 2023, showing aggregated sales by month.

Chapter 6

Conclusion and Future Scope

6.1 Conclusion

The SmartChain project presents an advanced solution for modern supply chain management by combining blockchain technology with AI-driven inventory management. It addresses key industry challenges such as lack of transparency, inefficiencies, and security vulnerabilities in traditional supply chains. Blockchain's decentralized and tamper-proof ledger ensures traceability and accountability at every stage of the supply chain, reducing the risks of fraud and errors, and improving overall transparency. On the other hand, the integration of AI enhances inventory management by optimizing demand forecasting, tracking, and real-time decision-making. In conclusion, SmartChain significantly enhances operational efficiency, security, and collaboration across the supply chain.

6.2 Future Scope

As SmartChain continues to evolve, we plan to expand its capabilities by developing a mobile application that mirrors the full functionality of the current Web3 full-stack platform. This mobile version will retain all the core features while introducing additional user-centric enhancements. Additionally, SmartChain's combination of decentralized finance (De-Fi) security with AI models ensures that it remains robust and scalable, positioning it as a leader in secure, AI-driven supply chain management.

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Acknowledgement

Success of a project like this involving high technical expertise, patience and massive support of guides, is possible when the team members work together. We take this opportunity to express our gratitude to those who have been instrumental in the successful completion of this project. We would like to show our appreciation to Mrs. Nupur Gaikawd for their tremendous support and help, without them this project would have reached nowhere. We would also like to thank our project coordinator Dr. Chhaya Pawar for providing us with regular inputs about documentation and project timeline. A big thanks to our HOD Dr. Kiruthika M. for all the encouragement given to our team. We would also like to thank our principal, Dr. S. M. Khot, and our college Fr. C. Rodrigues Institute of Technology, Vashi, for giving us the opportunity and the environment to learn and grow.

Reena Moncy(1022219) _______________ (Signature)

Parth Gopakumar(1022226) _______________ (Signature)

Savio Winson(1022247) _______________ (Signature)

Swastik Sharma(1022251) _______________ (Signature)

Appendix A: Timeline chart

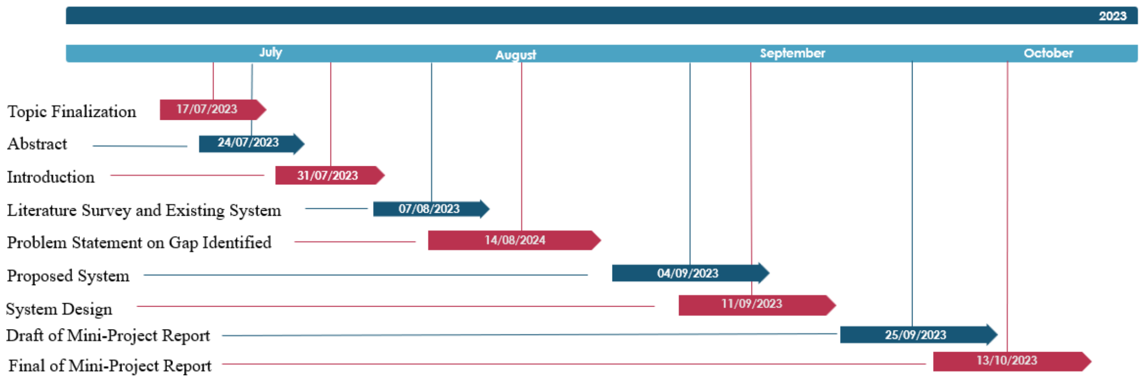

Figure 12: Timeline Chart 1

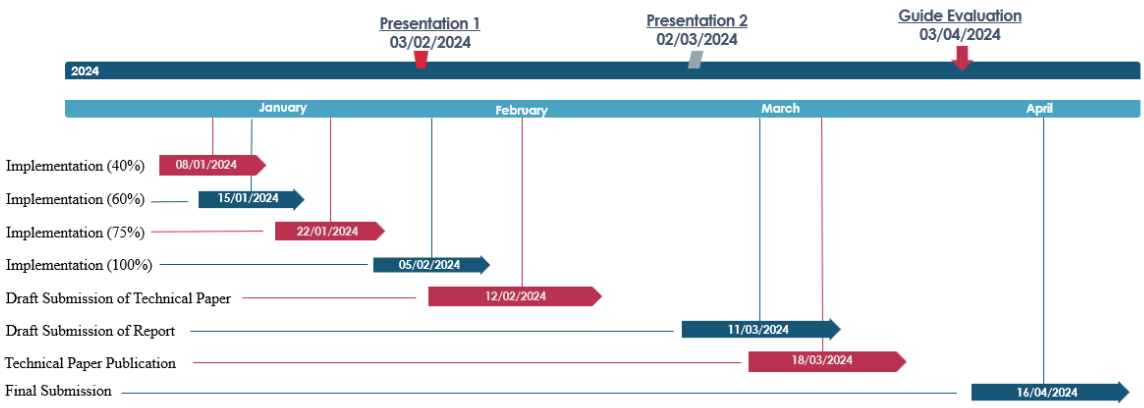

Figure 13: Timeline Chart 2

Appendix B: Publication Details

Outlook

3rd International Conference on Networks, Multimedia, and Information Technology (NMITCON-2025) : Submission (2061) has been created.

From Microsoft CMT <noreply@msr-cmt.org>
Date Sun 4/27/2025 5:47 PM
To SAVIO WINSON <1022247@COMP.fcrit.ac.in>

```
Hello,

The following submission has been created.

Track Name: NMITCON2025

Paper ID: 2061

Paper Title: SmartChain: Elevating Supply Chain Efficiency with AI and Blockchain

Abstract:
Traditional supply chain management systems are
hindered by persistent challenges, including widespread fraud,
operational inefficiencies, and a critical lack of transparency, all
of which contribute to substantial financial losses and delays
in goods delivery. To overcome these limitations, this paper
introduces SmartChain, an innovative framework that seam-
lessly integrates blockchain technology with artificial intelligence
(AI) to establish a robust, secure, and highly efficient supply
chain ecosystem. SmartChain leverages Ethereum-based smart
contracts to automate and secure critical supply chain processes,
such as procurement and logistics, ensuring that all movements
of goods are recorded in an immutable and transparent ledger
accessible to authorized stakeholders. Simultaneously, advanced
AI models enable precise demand forecasting, identify pat-
terns indicative of fraudulent behavior, and optimize inven-
tory management, reducing waste and enhancing operational
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Figure 14: Submission Confirmation for NMITCON-2025

This image shows the confirmation email received on April 27, 2025, for the submission of our paper titled "SmartChain: Elevating Supply Chain Efficiency with AI and Blockchain" to the 3rd International Conference on Networks, Multimedia, and Information Technology (NMITCON-2025). The paper, with submission ID 2061, was submitted by Savio Winson and addresses the challenges of traditional supply chain management by proposing an innovative framework that integrates blockchain technology and artificial intelligence. This submission highlights our team's efforts to enhance supply chain efficiency, security, and transparency, as detailed in the abstract.

Appendix C: Additional References

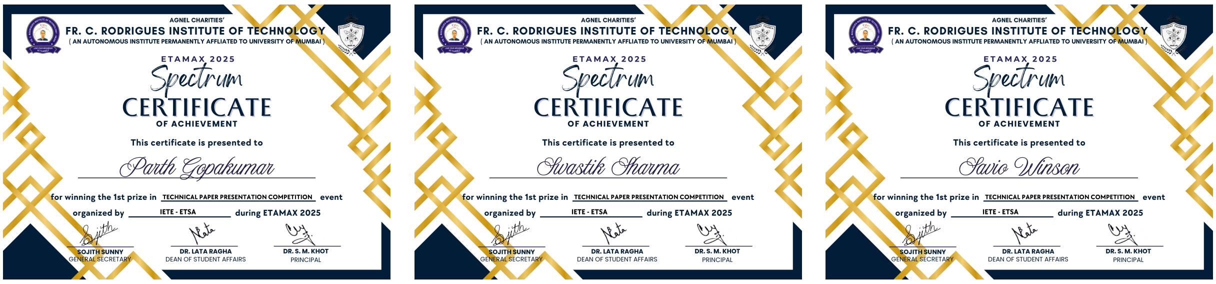

Figure 15: Certificates of Achievement for Etamax 2025

We are proud to announce that our team won the 1st prize in the Technical Paper Presentation Competition organized by IETE-ETSA during Etamax 2025, held at Fr. C. Rodrigues Institute of Technology, Vashi. This achievement highlights our successful presentation of the SmartChain project, recognizing our efforts in integrating blockchain and AI for supply chain management.